

Question

Heating the hydrated salt **A** of an active main group metal **M** under an air atmosphere begins to decompose at 471.2 K, forming compound **B** with a mass loss of 5.39% (theoretical mass fraction, same below; relative atomic masses are retained to two decimal places). Continued heating to 660.2 K produces compound **C** with a mass loss of 34.74%. Finally, decomposition begins at 968.7 K to form compound **D**, releasing a colorless and odorless gas **X**, with a mass loss of 26.32%. All mass loss percentages are based on **A**. Regarding **A** ~ **D**, which of the following statements is correct:

- A.** The chemical formula of **A** is $\text{B} \cdot 2\text{H}_2\text{O}$.
- B.** The chemical formula of **A** is $\text{B} \cdot \text{H}_2\text{O}$
- C.** **B** reacts with hydrochloric acid to produce a colorless and odorless acid.
- D.** **C** does not produce gas when reacting with hydrochloric acid
- E.** **D** does not react with water
- F.** **D** reacts with hydrochloric acid to yield a toxic salt.
- G.** Passing **X** into an aqueous solution of **B** yields **C**
- H.** The thermal decomposition of **B** into **C** can be regarded as the release of compound **E**, whose ideal point group is C_{2v} .
- I.** All other options are incorrect

Answer

Correct Answer: **H**

Detailed Explanation

Highly reactive main group metals indicate that **M** is an alkali metal or alkaline earth metal.

CHECKPOINT

1 PTS

M is an alkali metal or alkaline earth metal

Oxides of alkali metals or alkaline earth metals are difficult to decompose, so **X** is carbon dioxide, **C** is carbonate, and **D** is oxide.

CHECKPOINT

1 PTS

X is carbon dioxide, **C** is carbonate, and **D** is oxide

$$\frac{2M+16}{44.01} = \frac{0.3355}{0.2632}$$

or

$$\frac{M+16}{44.01} = \frac{0.3355}{0.2632}$$

Solving gives $M = 40.09$, **M** = Ca.

CHECKPOINT

1 PTS

M = Ca

A mass loss of 34.74% corresponds to 58.09, which is $\text{C}_2\text{H}_2\text{O}_2$ or $\text{C}_3\text{H}_6\text{O}$; the latter is reasonable, meaning the anion is acetate. Consequently, **A** has 0.5 water molecules of crystallization, so **A** = $(\text{CH}_3\text{CO}_2)_2\text{Ca} \cdot 0.5\text{H}_2\text{O}$, **B** = $(\text{CH}_3\text{CO}_2)_2\text{Ca}$, **C** = CaCO_3 , **D** = CaO .

CHECKPOINT

2 PTS

A = $(\text{CH}_3\text{CO}_2)_2\text{Ca} \cdot 0.5\text{H}_2\text{O}$, **B** = $(\text{CH}_3\text{CO}_2)_2\text{Ca}$, **C** = CaCO_3 , **D** = CaO

Options A and B are incorrect; **B** reacts with hydrochloric acid to produce the pungent-smelling acetic acid, so C is incorrect; **C** reacts with hydrochloric acid to produce carbon dioxide, so D is incorrect; **D** reacts with water, so E is incorrect; **D** reacts with hydrochloric acid to produce non-toxic calcium chloride, so F is incorrect; **B** cannot react with CO_2 to form **C**, so G is incorrect.

B decomposes into **C** and loses acetone, which belongs to the C_{2v} point group, so H is correct.

CHECKPOINT

1 PTS

E is acetone